

Chapter 1: Getting Started and Prerequisites

Welcome to the Cloud-Native Retrieval-Augmented Generation deployment guide.

Before beginning installation, ensure your environment meets the minimum hardware and software prerequisites:

- Python 3.10 or higher installed with pip and virtualenv support.
- At least 8GB of system RAM (16GB recommended for batch embedding workloads).
- CUDA-compatible GPU with minimum 4GB VRAM (optional, CPU execution supported via faiss-cpu).

Installation Steps:

1. Clone the repository from the centralized git hosting platform.
2. Initialize and activate an isolated Python virtual environment.
3. Install required runtime dependencies using the requirements lockfile.
4. Verify environment readiness by running the automated self-test diagnostic suite.

Chapter 2: Text Ingestion and Sentence Chunking Architecture

Document chunking represents the cornerstone of effective semantic retrieval.

When documents are poorly partitioned, LLMs suffer from incomplete context or broken semantic phrases.

Key Rules for Chunk Configuration:

- Target Chunk Size: Recommended between 500 and 1000 characters. Chunks smaller than 300 characters lack semantic context, while chunks larger than 1500 characters dilute vector precision.
- Overlap Ratio: Maintain 10-15% character overlap between adjacent chunks to guarantee sentence continuity.
- Abbreviation Safety: Ensure abbreviations such as Dr., Prof., e.g., and i.e. do not prematurely trigger sentence splits.
- Word Boundary Preservation: Never slice words mid-token during character fallback.

Chapter 3: Maintenance, Monitoring, and Troubleshooting

Routine maintenance ensures vector store indexes remain optimized for low-latency queries.

Monitoring Recommendations:

- Track average search latency per query (target: < 20ms).
- Monitor index memory footprint (FAISS FlatIP consumes ~1.5KB per indexed 384-d vector).
- Audit chunk verification logs for empty or oversized chunks prior to embedding ingestion.

Troubleshooting Common Issues:

If retrieval scores drop below the 0.35 confidence threshold, re-evaluate chunk sizes and document parsing quality.

Ensure that scanned or image-only documents undergo Optical Character Recognition (OCR) before ingestion.